Dr. Thyagarajulu Gollapalli

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Research Interests

Numerical Modelling, Machine/Deep Learning, High Performance Computing, GIS, Megathrust Seismicity, Groundwater Modelling, Geophysical Data Acquisition, Processing, and Interpretation

Current Employment

Research Officer/Software Engineer — **Canberra, ACT**
Australian National University — *Jan. 2026 - Present*

- Open-source developer for the Underworld codebase, contributing to core features and user-facing interfaces while enhancing modelling capabilities through integration of geological and geophysical processes. Leveraged AI-assisted tools to accelerate development, testing, and debugging, improving code quality, research productivity, and reproducibility.

Work Experience

Assistant Lecturer (Maternity Cover) — **Melbourne, VIC**
Monash University — *Jun. 2025 - Dec. 2025*

- **EAE3032 (Undergraduate) – Geophysical Interpretations in geoscience, geography and environmental science**
- **EAE5053 (Postgraduate) – Advanced Spatial Analysis and Modelling**

Senior Research Officer/Software Engineer — **Melbourne, VIC**
Monash University and AuScope — *Aug. 2023 - Jun. 2025*

- Responsible for high-quality research computing services for the AuScope Simulation, Analysis and Modelling (SAM) node, including core development of the Underworld codebase and user-facing interfaces for the geodynamics community; led testing and debugging, prepared user documentation, and organised training workshops.

Imaging Geophysicist — **Perth, WA**
CGG — *Nov. 2022 - Jun. 2023*

- Applied advanced subsurface imaging to resolve complex geology and enhance image quality; designed, executed, and monitored end-to-end seismic processing with rigorous QC throughout; managed schedules and HPC resources to manage deliverables on time while communicating progress and risks to stakeholders.

Education Background

Monash University & Indian Institute of Technology (IIT) Bombay — **Melbourne, VIC**
Doctor of Philosophy, Computational Geodynamics & Geophysics — *Apr. 2018 - Nov. 2022*
Thesis: Unravelling tectonic coupling and loading along Sunda margin through 3-D regional numerical modelling

Advisors: Fabio A. Capitanio, Monash University & M. Radhakrishna, IIT Bombay

Indian Institute of Science (IISc) — **Bengaluru, India**
*Bachelor of Science, **Major:** Mathematics,* — *Aug. 2011 - May 2015*
***Minor:** Earth & Environmental Science*

Thesis: One-dimensional totally asymmetric simple exclusion process
Advisor: Arvind Ayyer, Department of Mathematics

Research Experience

Department of Earth Sciences, IIT Bombay **Mumbai, India**
Junior Research Fellow in Gravity & Magnetics Lab *May 2017 - Apr. 2018*

Research: Structure and Rheological characteristics of the lithosphere below the Northern Indian Ocean based on the Gravity-Geoid modeling.
Advisor: *M. Radhakrishna*

Centre for Earth Sciences, IISc **Bengaluru, India**
Project Assistant in Computational Geodynamics Lab *Aug. 2015 - Mar. 2017*

Research: Investigating Indian ocean geoid low.
Advisor: *Attreyee Ghosh*

Centre for Earth Sciences, IISc **Bengaluru, India**
Undergraduate Research Assistant in Thin section Lab *May 2014 - Jul. 2014*

Research: Delineating metamorphism effect in rock minerals along chilled margins.
Advisor: *Kusala Rajendran*

Centre for Earth Sciences, IISc **Bengaluru, India**
Undergraduate Research Assistant in Mass Spectrometery Lab *May 2013 - Jul. 2013*

Research: Latitudinal dependency of carbonates from dental enamel.
Advisor: *Prosenjit Ghosh*

Department of Mathematics, IISc **Bengaluru, India**
Undergraduate Summer Intern *May 2012 - Jul. 2012*

Research: Group theory reading and discussion sessions.
Advisor: *Kaushal Verma*

Publications

1. Khalil, H., **Gollapalli, T**., Capitanio, F.A., Knight, B., & Brune, S., How Far-Field Rotational Motion Drives Rift Evolution, *Earth and Planetary Science Letters*, (Submitted).
2. Capitanio, F.A., **Gollapalli, T**., Munukutla, R., Graciosa, J.C., Zuhair, M., Beall, A., & Dal Zilio, L., Bridging the gap between subduction dynamics and the long-term strength of megathrusts, *Nature Communications*, DOI: https://doi.org/10.1038/s41467-025-65824-7, 2025.
3. Moresi, L., Mansour, J., Giordani, J., Knepley, M., Knight, B., Graciosa, J. C., **Gollapalli, T**., Lu, N., & Beucher, R., Underworld3: Mathematically Self-Describing Modelling in Python for Desktop, HPC and Cloud, *JOSS*, DOI: 10.21105/joss.07831, 2025.
4. Graciosa, J. C., Capitanio, F. A., Beall, A., Hargreaves, M., **Gollapalli, T**., Tang, T., & Zuhair, M., Testing Driving Mechanisms of Megathrust Seismicity With Explainable Artificial Intelligence, *JGR Solid Earth*, https://doi.org/10.1029/2024JB028774, 2025.
5. Zuhair, M., **Gollapalli, T**., Capitanio, F.A., Betts, P.G., & Graciosa, J.C., The role of slab step on tectonic loading along subduction zones: inferences on the seismotectonics of the Sunda convergent margin, *Tectonics*, https://doi.org/10.1029/2022TC007242, 2022.
6. Ghosh, A., **Gollapalli, T.**, & Steinberger, B., The importance of upper mantle heterogeneity in generating the Indian Ocean geoid low, *Geophysical Research Letters*, https://doi.org/10.1002/2017GL075392, 2017.

Conference Posters & Presentations (selected)

1. **European Geosciences Union General Assembly,** Oral Session, 23-27 May 2022, "Megathrust Seismicity Through the Lens of Explainable Artificial Intelligence".
2. **American Geophysical Union Fall Meeting,** Oral Session, 13-17 Dec 2021, "Subduction dynamics and Tectonic coupling along seismically active margin: the relevance of Trench-parallel forces".

3. **American Geophysical Union Fall Meeting,** Poster Session, 9-13 Dec 2019, "Megathrust Seismotectonics of the Southeast Asian Convergent Margin: Insights from Three-Dimensional Spherical Geodynamic Modelling".

Blog Posts

- Unlocking the secrets of great earthquakes with deep learning, 2022 (link)
- Advanced geodynamic models of giant earthquakes, 2019 (link)

Skills

- **Technical:**
  Geodynamic Codes: Underworld, CitcomS, Litmod-3D, ASPECT, GPlates.
  Advanced: Python, Matplotlib, Cartopy, Scikit, Plotly, Matlab, AWK, Shell script, Generic Mapping Tools (GMT), Paraview, Microsoft Tools, Git.
  Intermediate: QGIS, Surfer, C, Docker, PyCharm, PyVista.
  Basic: Fortran, Julia, R, PyTorch, Seaborn.
- **Instruments:** Gravimeter (CG-5 Autograv), Magnetometer (GST-19M Proton Precession), Isotope Ratio Mass Spectrometer (IRMS MAT 253), Thin section & Petrology Lab Instruments.
- **Core:** Data Analysis, Data Visualisation, ML, DL, HPC, GIS, Problem Solving, Leadership, Adaptability, Relationship Building.

Scholarships & Awards

- **Postgraduate Publications Award (PPA) $5,151 AUD,** Apr-Jun 2022, Monash University.
- **AGU Fall Meeting Student Travel Grant $1,000 USD,** 13-17 Dec 2021.
- **Graduate Research Completion Award (GRCA) ∼$5,000 AUD,** Dec 2021 - Jan 2022, Monash University
- **Monash Graduate Scholarship $27,353 AUD/annum,** 2018-2021, Monash University.
- **Faculty of Science Dean's International Postgraduate Research Scholarship $176,297 AUD,** 2018-2022, Monash University.
- **Junior Research Fellow Scholarship,** 2017-2018, Ministry of Human Resource Development (MHRD), India.
- **INSPIRE Scholarship for Higher Education,** 2011-2015, Department of Science and Technology (DST), India.

Teaching Assistant Experience

- **Physics of the Solid Earth, EAE3532, Teaching Assistant,** 2019, 10 students, teaching, grading, reviews.
- **Earth, Atmosphere and Environment 1, EAE1011, Teaching Assistant,** 2020, ∼300 students, teaching, grading, reviews.

Outreach & Service

- **Open Day Demonstrator,** Monash, 2022 & IISc, 2016, 2017
- **Protolith Volunteer,** Biennial festival, IIT Bombay, 2017
- **Member of EAE Outreach Team,** Monash University, 2019

Standardised Tests

- Indian Institute of Technology Joint Entrance Examination (IIT-JEE), 2011, ranked in top 1% of Indian applying for undergraduate study in 2011.
- Graduate Aptitude Test in Engineering (GATE), 2016, ranked in top 20% in Geology & Geophysics stream.

Workshops

- Underworld workshop at Curtin University, 2025.
- Underworld workshop at AuScope HQ, 2024.
- Underworld workshop at University of Syndey, 2023.
- The Victorian Universities Symposium on Computational Geodynamics, 3 Nov 2021. Organised by University of Melbourne.
- Graduate Research Conference, 19 Nov 2019. Organised by Monash University.
- Quantitative Seismology and Microseismicity using Geophysical and Geochemical Techniques, 2017. Organised by CEaS, IISc and Shell Technology Centre, Bengarulu.
- Vijyoshi Camp (annual national science camp), 26-27 Nov 2011. Organised by IISc, Bengarulu.

Interests

Marathon Runner, Cycling, Badminton

- *Nike Melbourne Marathon*, 2021, 42.2 km, Time-03:57:26
- *Wings for Life World Run*, 2022, 18 km, Time-01:43:11
- *Great Ocean Road Marathon*, 2024, 44 km, Time-04:03:32